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Computer Organization and Architecture
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Chapter 10

Computer Arithmetic

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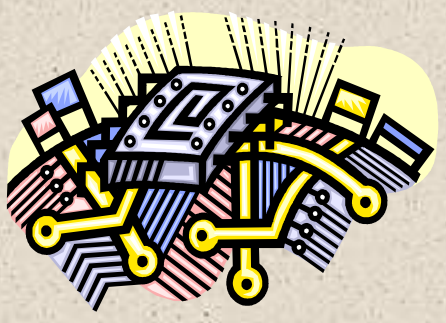
Arithmetic & Logic Unit (ALU)

- Part of the computer that actually performs arithmetic and logical operations on data
- All of the other elements of the computer system are there mainly to bring data into the ALU for it to process and then to take the results back out
- Based on the use of simple digital logic devices that can store binary digits and perform simple Boolean logic operations



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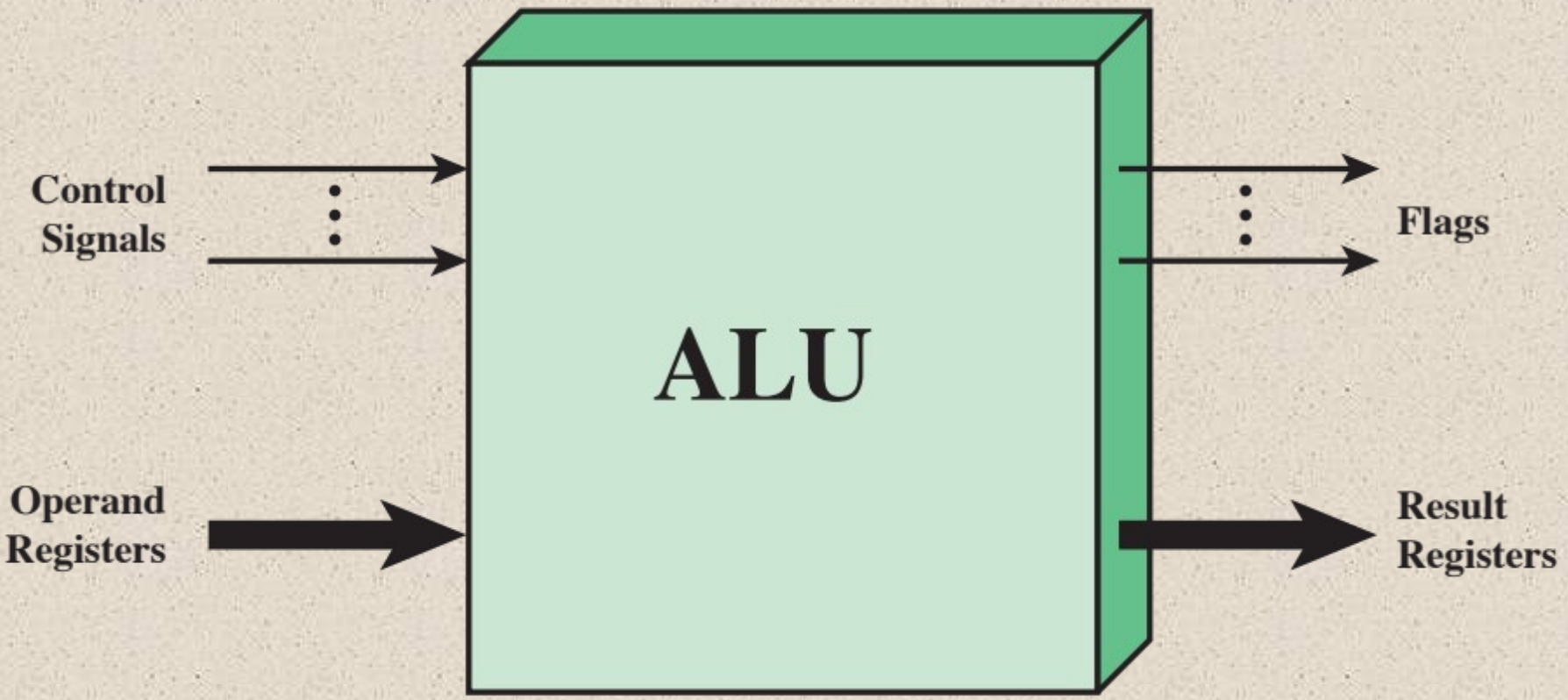

Figure 10.1 ALU Inputs and Outputs

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Integer Representation

- In the binary number system arbitrary numbers can be represented with:
  - The digits zero and one
  - The minus sign (for negative numbers)
  - The period, or ***radix point*** (for numbers with a fractional component)
- For purposes of computer storage and processing we do not have the benefit of special symbols for the minus sign and radix point
- Only binary digits (0,1) may be used to represent numbers

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Sign-Magnitude Representation

There are several alternative conventions used to represent negative as well as positive integers	•All of these alternatives involve treating the most significant (leftmost) bit in the word as a sign bit •If the sign bit is 0 the number is positive •If the sign bit is 1 the number is negative
Sign-magnitude representation is the simplest form that employs a sign bit	
Drawbacks:	•Addition and subtraction require a consideration of both the signs of the numbers and their relative magnitudes to carry out the required operation •There are two representations of 0
Because of these drawbacks, sign-magnitude representation is rarely used in implementing the integer portion of the ALU	

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Table 10.1
Characteristics of Twos Complement Representation and Arithmetic

Range	-2_{n-1} through $2_{n-1} - 1$
Number of Representations of Zero	One
Negation	Take the Boolean complement of each bit of the corresponding positive number, then add 1 to the resulting bit pattern viewed as an unsigned integer.
Expansion of Bit Length	Add additional bit positions to the left and fill in with the value of the original sign bit.
Overflow Rule	If two numbers with the same sign (both positive or both negative) are added, then overflow occurs if and only if the result has the opposite sign.
Subtraction Rule	To subtract *B* from *A*, take the twos complement of *B* and add it to *A*.

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Table 10.2
Alternative Representations for 4-Bit Integers

Decimal Representation	Sign-Magnitude Representation	Twos Complement Representation	Biased Representation
+8	—	—	1111
+7	0111	0111	1110
+6	0110	0110	1101
+5	0101	0101	1100
+4	0100	0100	1011
+3	0011	0011	1010
+2	0010	0010	1001
+1	0001	0001	1000
+0	0000	0000	0111
–0	1000	—	—
–1	1001	1111	0110
–2	1010	1110	0101
–3	1011	1101	0100
–4	1100	1100	0011
–5	1101	1011	0010
–6	1110	1010	0001
–7	1111	1001	0000
–8	—	1000	—

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–128	64	32	16	8	4	2	1

(a) An eight-position two's complement value box

–128	64	32	16	8	4	2	1
1	0	0	0	0	0	1	1

–128 +2 +1 = –125

(b) Convert binary 10000011 to decimal

–128	64	32	16	8	4	2	1
1	0	0	0	1	0	0	0

–120 = –128 +8

(c) Convert decimal –120 to binary

Figure 10.2 Use of a Value Box for Conversion Between Twos Complement Binary and Decimal

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Range Extension

- Range of numbers that can be expressed is extended by increasing the bit length
- In sign-magnitude notation this is accomplished by moving the sign bit to the new leftmost position and fill in with zeros
- This procedure will not work for twos complement negative integers
  - Rule is to move the sign bit to the new leftmost position and fill in with copies of the sign bit
  - For positive numbers, fill in with zeros, and for negative numbers, fill in with ones
  - This is called *sign extension*

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Fixed-Point Representation

The radix point (binary point) is fixed and assumed to be to the right of the rightmost digit

Programmer can use the same representation for binary fractions by scaling the numbers so that the binary point is implicitly positioned at some other location

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Negation

- Twos complement operation
  - Take the Boolean complement of each bit of the integer (including the sign bit)
  - Treating the result as an unsigned binary integer, add 1

```
                +18 = 00010010 (twos complement)
bitwise complement = 11101101
                     +        1
                      11101110 = -18
```

- The negative of the negative of that number is itself:

```
                -18 =  11101110 (twos complement)
bitwise complement =  00010001
                      +        1
                        00010010 = +18
```

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Negation Special Case 1

0 = 00000000 (twos complement)

Bitwise complement = 11111111

Add 1 to LSB + 1

Result 100000000

Overflow is ignored, so:

- 0 = 0

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Negation Special Case 2

-128 = 10000000 (twos complement)

Bitwise complement = 01111111

Add 1 to LSB + 1

Result 10000000

So:

-(-128) = -128 X

Monitor MSB (sign bit)

It should change during negation

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1001 = –7 +0101 = 5 1110 = –2	1100 = –4 +0100 = 4 10000 = 0
(a) (–7) + (+5)	(b) (–4) + (+4)
0011 = 3 +0100 = 4 0111 = 7	1100 = –4 +1111 = –1 11011 = –5
(c) (+3) + (+4)	(d) (–4) + (–1)
0101 = 5 +0100 = 4 1001 = Overflow	1001 = –7 +1010 = –6 10011 = Overflow
(e) (+5) + (+4)	(f) (–7) + (–6)

Figure 10.3 Addition of Numbers in Twos Complement Representation

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Overflow

Rule

OVERFLOW RULE:

If two numbers are added, and they are both positive or both negative, then overflow occurs if and only if the result has the opposite sign.

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Subtraction

Rule

SUBTRACTION RULE:

To subtract one number (subtrahend) from another (minuend), take the twos complement (negation) of the subtrahend and add it to the minuend.

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0010 = 2 +1001 = –7 1011 = –5	0101 = 5 +1110 = –2 10011 = 3
(a) M = 2 = 0010 S = 7 = 0111 –S = 1001	(b) M = 5 = 0101 S = 2 = 0010 –S = 1110
1011 = –5 +1110 = –2 11001 = –7	0101 = 5 +0010 = 2 0111 = 7
(c) M =–5 = 1011 S = 2 = 0010 –S = 1110	(d) M = 5 = 0101 S =–2 = 1110 –S = 0010
0111 = 7 +0111 = 7 1110 = Overflow	1010 = –6 +1100 = –4 10110 = Overflow
(e) M = 7 = 0111 S = –7 = 1001 –S = 0111	(f) M = –6 = 1010 S = 4 = 0100 –S = 1100

Figure 10.4 Subtraction of Numbers in Twos Complement Representation (M – S)

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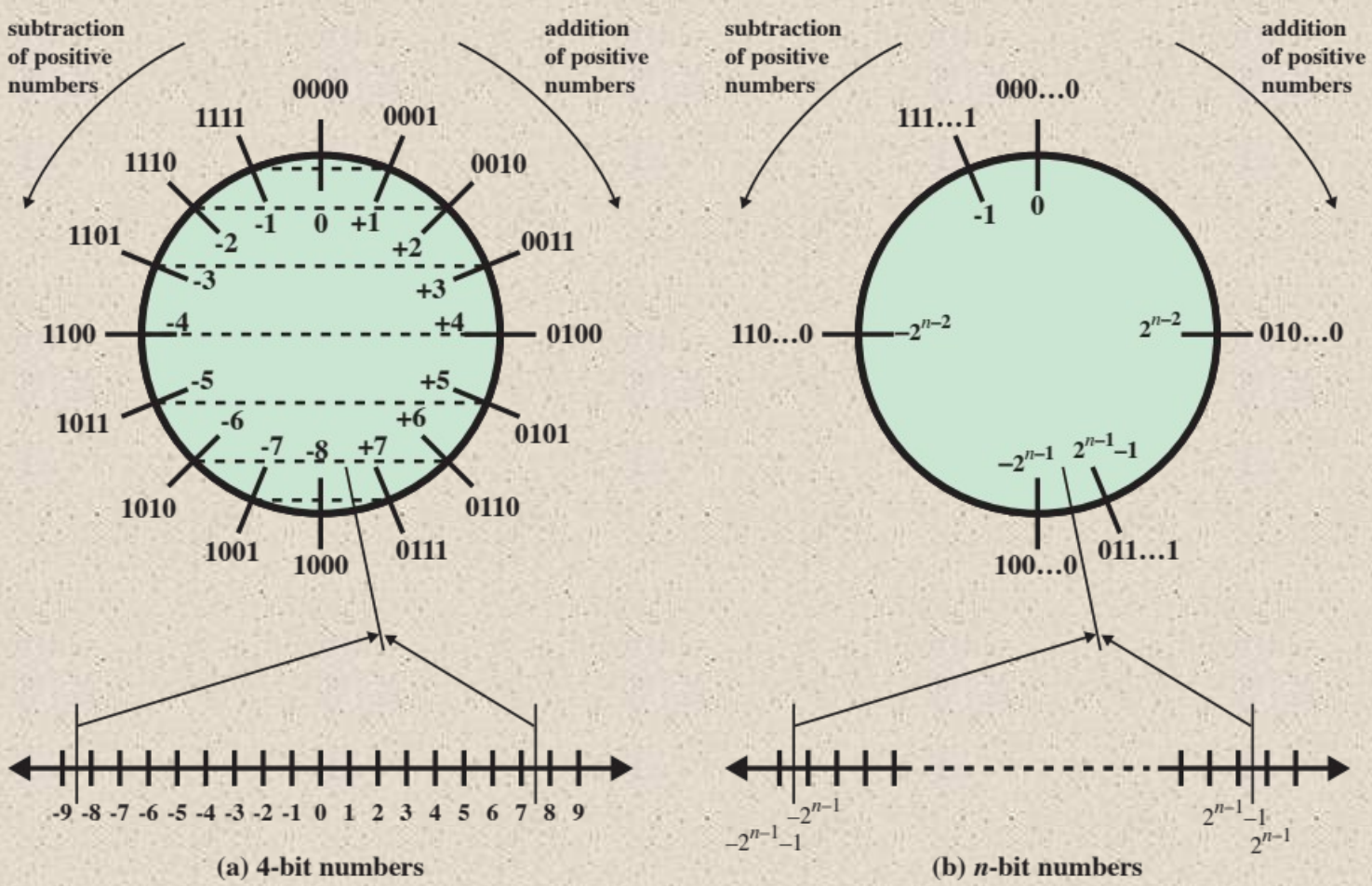

Figure 10.5 Geometric Depiction of Twos Complement Integers

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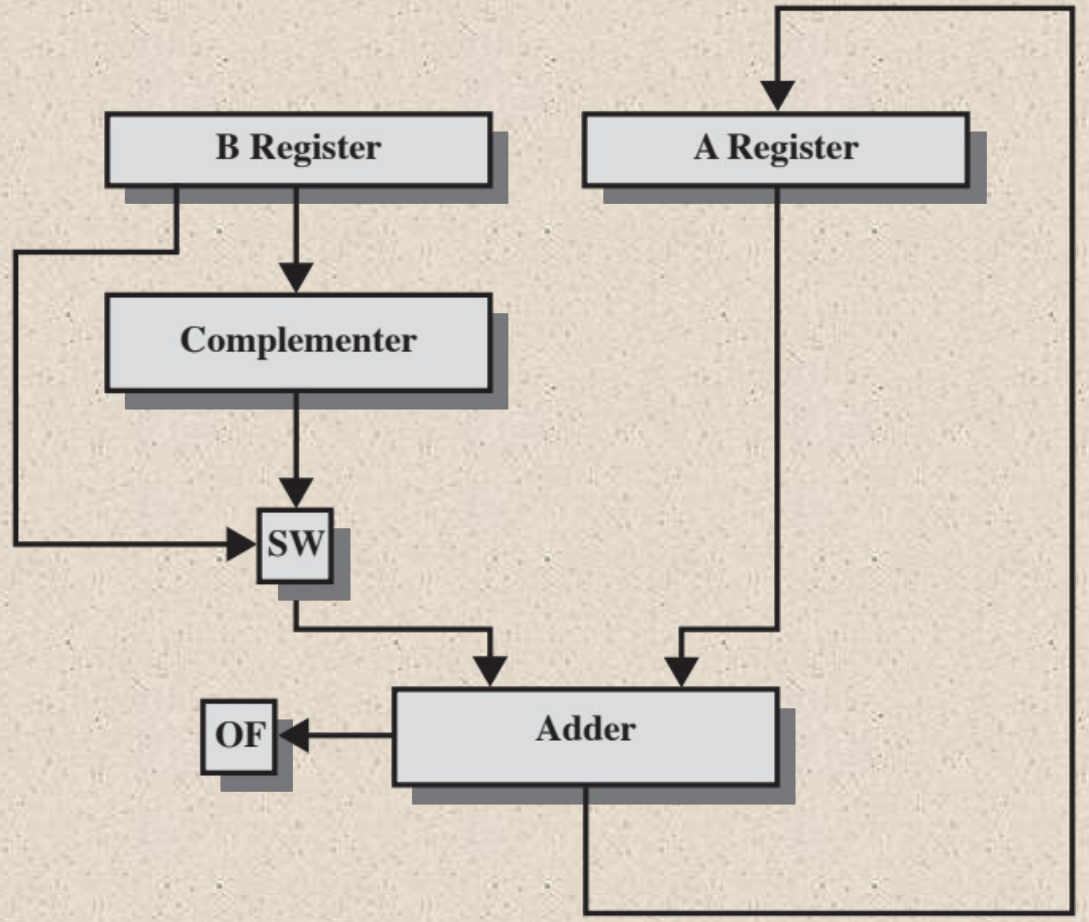

OF = overflow bit
SW = Switch (select addition or subtraction)

Figure 10.6 Block Diagram of Hardware for Addition and Subtraction

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```
    1011      Multiplicand (11)
  × 1101      Multiplier (13)
  ------
    1011  }
   0000   }   Partial products
  1011    }
 1011     }
--------
10001111      Product (143)
```

Figure 10.7 Multiplication of Unsigned Binary Integers

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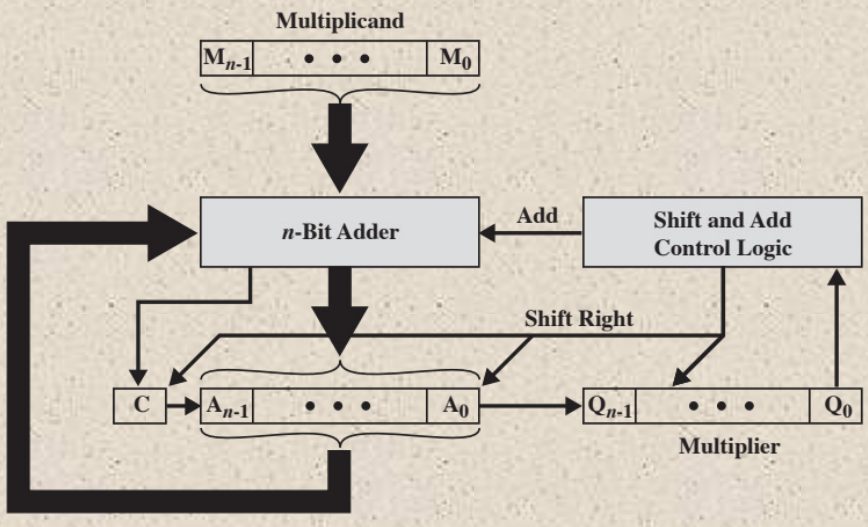

(a) Block Diagram

C	A	Q	M		
0	0000	1101	1011	Initial Values	
0	1011	1101	1011	Add	First Cycle
0	0101	1110	1011	Shift	
0	0010	1111	1011	Shift	Second Cycle
0	1101	1111	1011	Add	Third Cycle
0	0110	1111	1011	Shift	
1	0001	1111	1011	Add	Fourth Cycle
0	1000	1111	1011	Shift	

(b) Example from Figure 9.7 (product in A, Q)

Figure 10.8 Hardware Implementation of Unsigned Binary Multiplication

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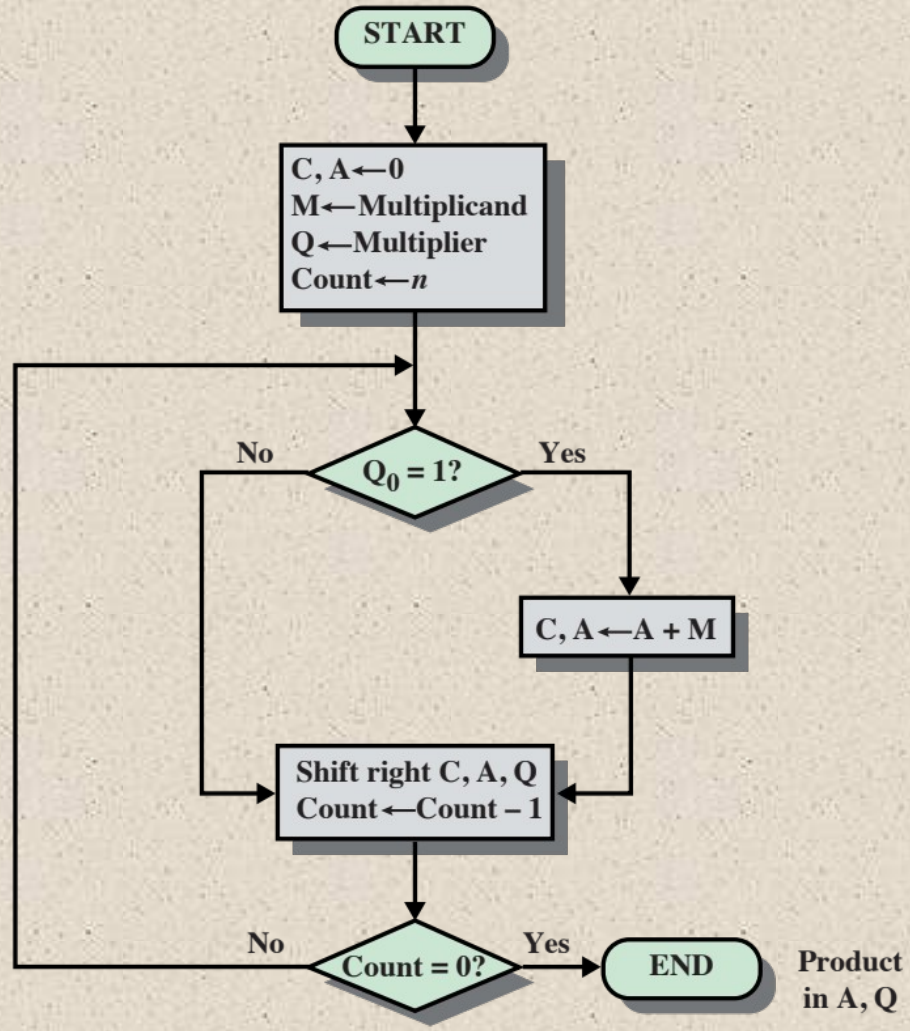

Figure 10.9 Flowchart for Unsigned Binary Multiplication

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```
    1011
   ×1101
--------
00001011   1011 × 1 × 2^0
00000000   1011 × 0 × 2^1
00101100   1011 × 1 × 2^2
01011000   1011 × 1 × 2^3
--------
10001111
```

Figure 10.10 Multiplication of Two Unsigned 4-Bit Integers Yielding an 8-Bit Result

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1001 (9)	1001 (−7)
×0011 (3)	×0011 (3)
00001001 1001×2^0	11111001 $(-7) \times 2^0 = (-7)$
00010010 1001×2^1	11110010 $(-7) \times 2^1 = (-14)$
00011011 (27)	11101011 (−21)
(a) Unsigned integers	(b) Twos complement integers

Figure 10.11 Comparison of Multiplication of Unsigned and Twos Complement Integers

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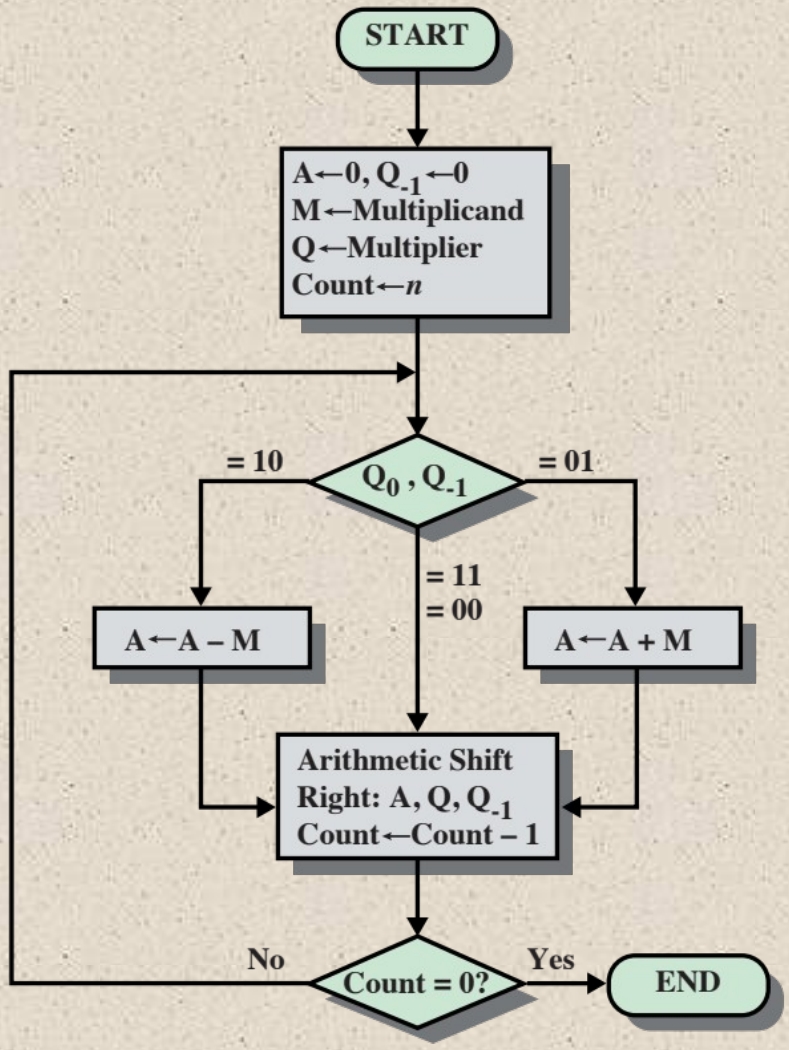

Figure 10.12 Booth's Algorithm for Twos Complement Multiplication

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A	Q	Q_{-1}	M		
0000	0011	0	0111	Initial Values	
1001	0011	0	0111	$A \leftarrow A - M$	First Cycle
1100	1001	1	0111	Shift	
1110	0100	1	0111	Shift	Second Cycle
0101	0100	1	0111	$A \leftarrow A + M$	Third Cycle
0010	1010	0	0111	Shift	
0001	0101	0	0111	Shift	Fourth Cycle

Figure 10.13 Example of Booth's Algorithm (7× 3)

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```
    0111
   ×0011      (0)
11111001      1–0
0000000       1–1
000111        0–1
00010101      (21)
```

(a) (7) × (3) = (21)

```
    0111
   ×1101      (0)
11111001      1–0
0000111       0–1
111001        1–0
11101011      (–21)
```

(b) (7) × (–3) = (–21)

```
    1001
   ×0011      (0)
00000111      1–0
0000000       1–1
111001        0–1
11101011      (–21)
```

(c) (–7) × (3) = (–21)

```
    1001
   ×1101      (0)
00000111      1–0
1111001       0–1
000111        1–0
00010101      (21)
```

(d) (–7) × (–3) = (21)

Figure 10.14 Examples Using Booth's Algorithm

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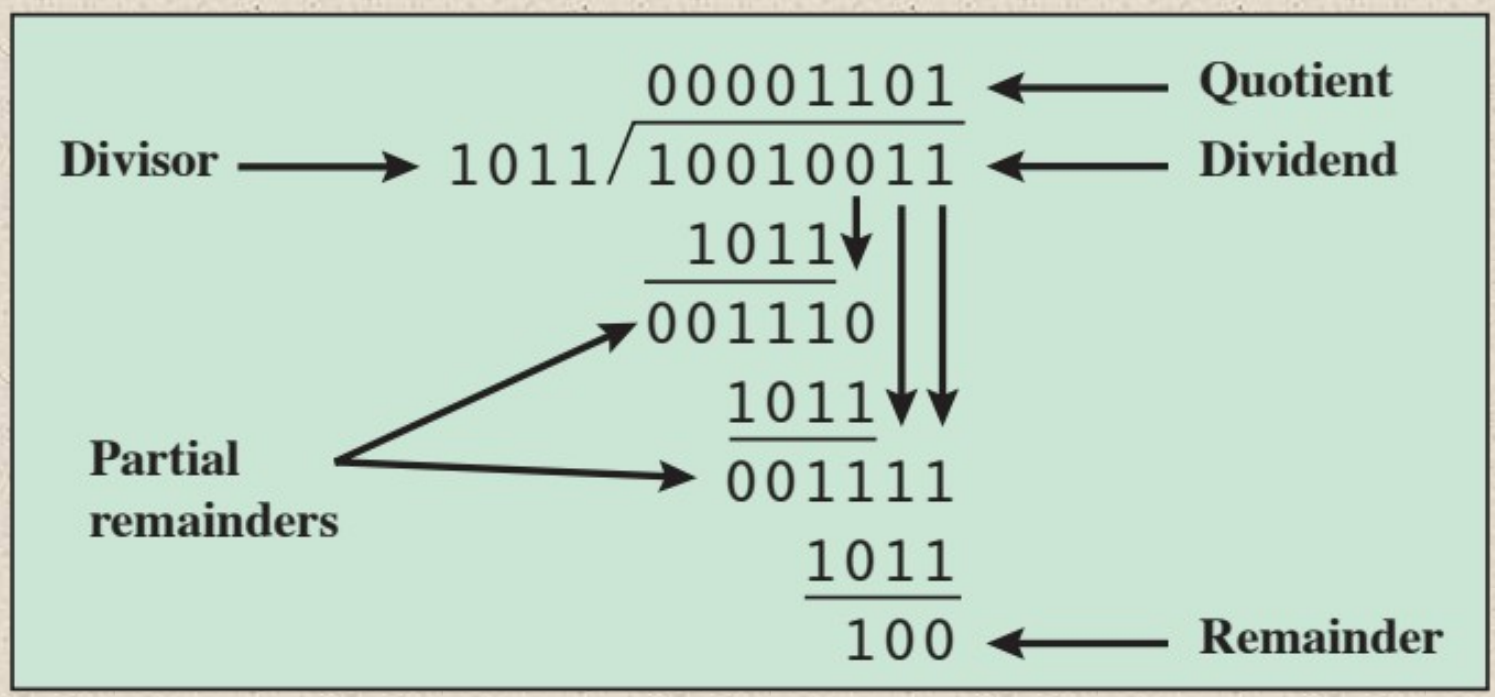

Figure 10.15 Example of Division of Unsigned Binary Integers

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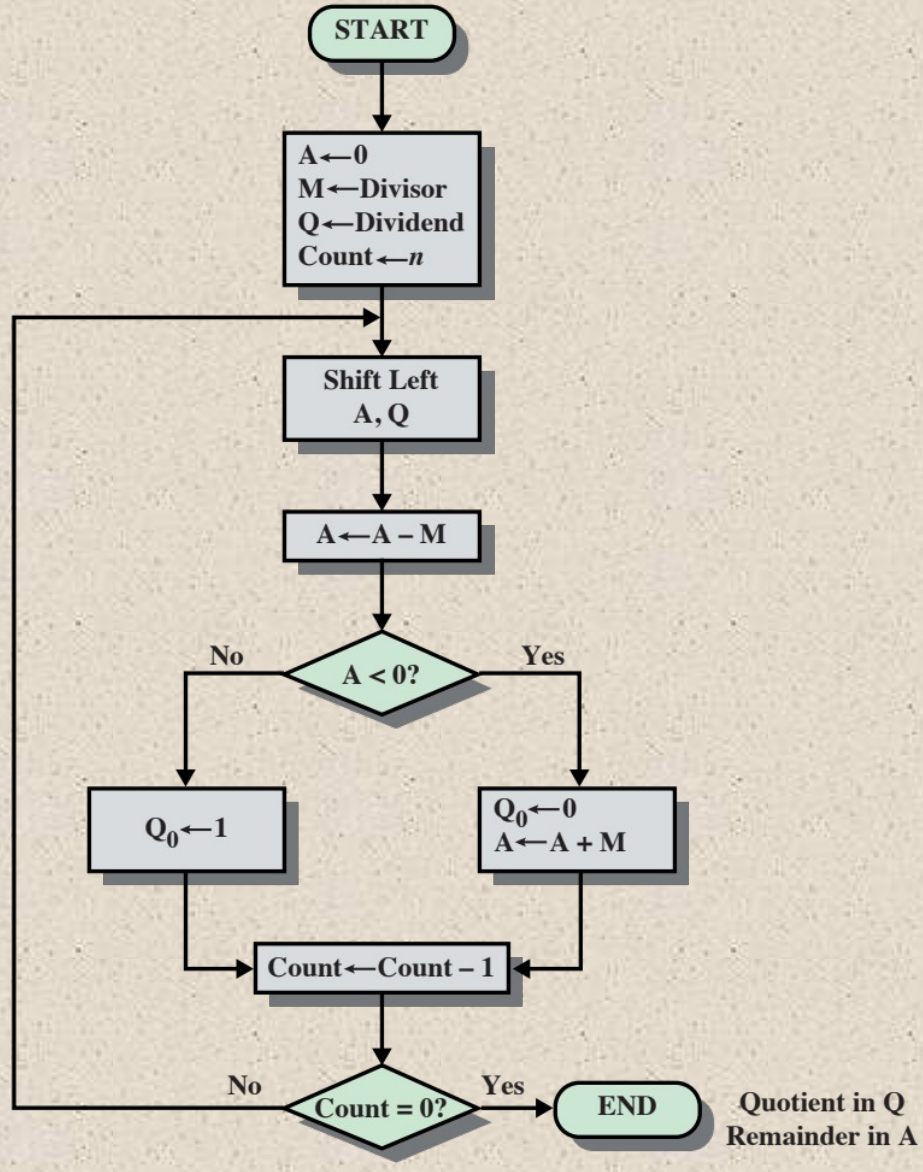

Figure 10.16 Flowchart for Unsigned Binary Division

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A	Q	
0000	0111	Initial value
0000	1110	Shift
1101		Use twos complement of 0011 for subtraction
1101		Subtract
0000	1110	Restore, set $Q_0 = 0$
0001	1100	Shift
1101		
1110		Subtract
0001	1100	Restore, set $Q_0 = 0$
0011	1000	Shift
1101		
0000	1001	Subtract, set $Q_0 = 1$
0001	0010	Shift
1101		
1110		Subtract
0001	0010	Restore, set $Q_0 = 0$

Figure 10.17 Example of Restoring Twos Complement Division (7/3)

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Floating-Point Representation

Principles

- With a fixed-point notation it is possible to represent a range of positive and negative integers centered on or near 0
- By assuming a fixed binary or radix point, this format allows the representation of numbers with a fractional component as well
- Limitations:
  - Very large numbers cannot be represented nor can very small fractions
  - The fractional part of the quotient in a division of two large numbers could be lost

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(a) Format

$$1.1010001 \times 2^{10100} = 0\ 10010011\ 10100010000000000000000 = 1.6328125 \times 2^{20}$$
$$-1.1010001 \times 2^{10100} = 1\ 10010011\ 10100010000000000000000 = -1.6328125 \times 2^{20}$$
$$1.1010001 \times 2^{-10100} = 0\ 01101011\ 10100010000000000000000 = 1.6328125 \times 2^{-20}$$
$$-1.1010001 \times 2^{-10100} = 1\ 01101011\ 10100010000000000000000 = -1.6328125 \times 2^{-20}$$

(b) Examples

Figure 10.18 Typical 32-Bit Floating-Point Format

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Floating-Point

Significand

- The final portion of the word
- Any floating-point number can be expressed in many ways

The following are equivalent, where the significand is expressed in binary form:

$$0.110 * 2^5$$
$$110 * 2^2$$
$$0.0110 * 2^6$$

- *Normal number*
  - The most significant digit of the significand is nonzero

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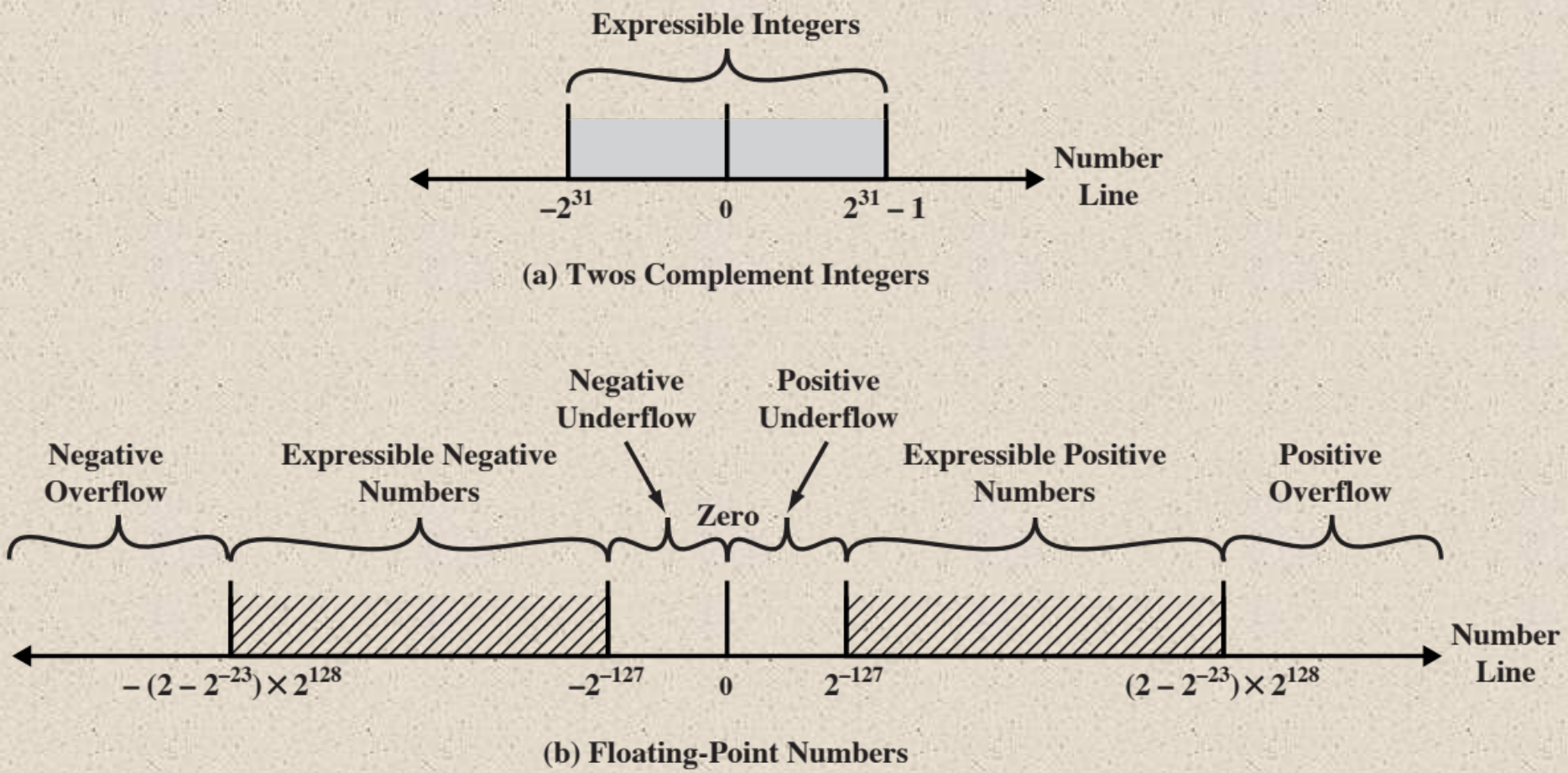

Figure 10.19 Expressible Numbers in Typical 32-Bit Formats

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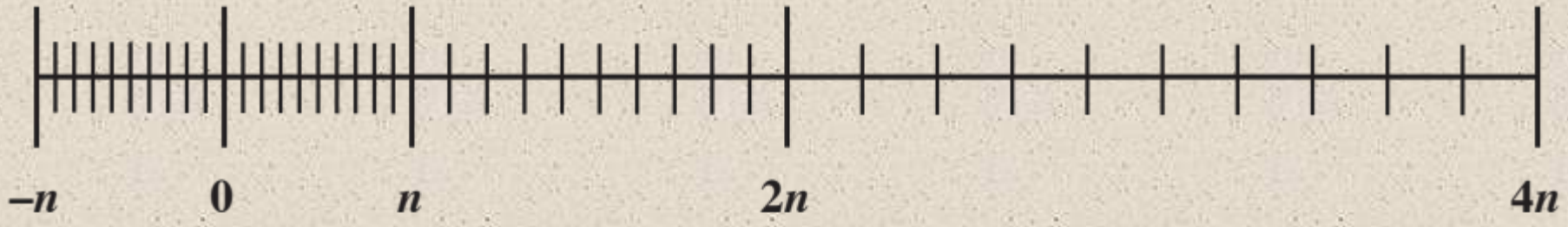

Figure 10.20 Density of Floating-Point Numbers

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IEEE Standard 754

Most important floating-point representation is defined

Standard was developed to facilitate the portability of programs from one processor to another and to encourage the development of sophisticated, numerically oriented programs

Standard has been widely adopted and is used on virtually all contemporary processors and arithmetic coprocessors

IEEE 754-2008 covers both binary and decimal floating-point representations

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IEEE 754-2008

- Defines the following different types of floating-point formats:
- Arithmetic format
  - All the mandatory operations defined by the standard are supported by the format. The format may be used to represent floating-point operands or results for the operations described in the standard.
- Basic format
  - This format covers five floating-point representations, three binary and two decimal, whose encodings are specified by the standard, and which can be used for arithmetic. At least one of the basic formats is implemented in any conforming implementation.
- Interchange format
  - A fully specified, fixed-length binary encoding that allows data interchange between different platforms and that can be used for storage.

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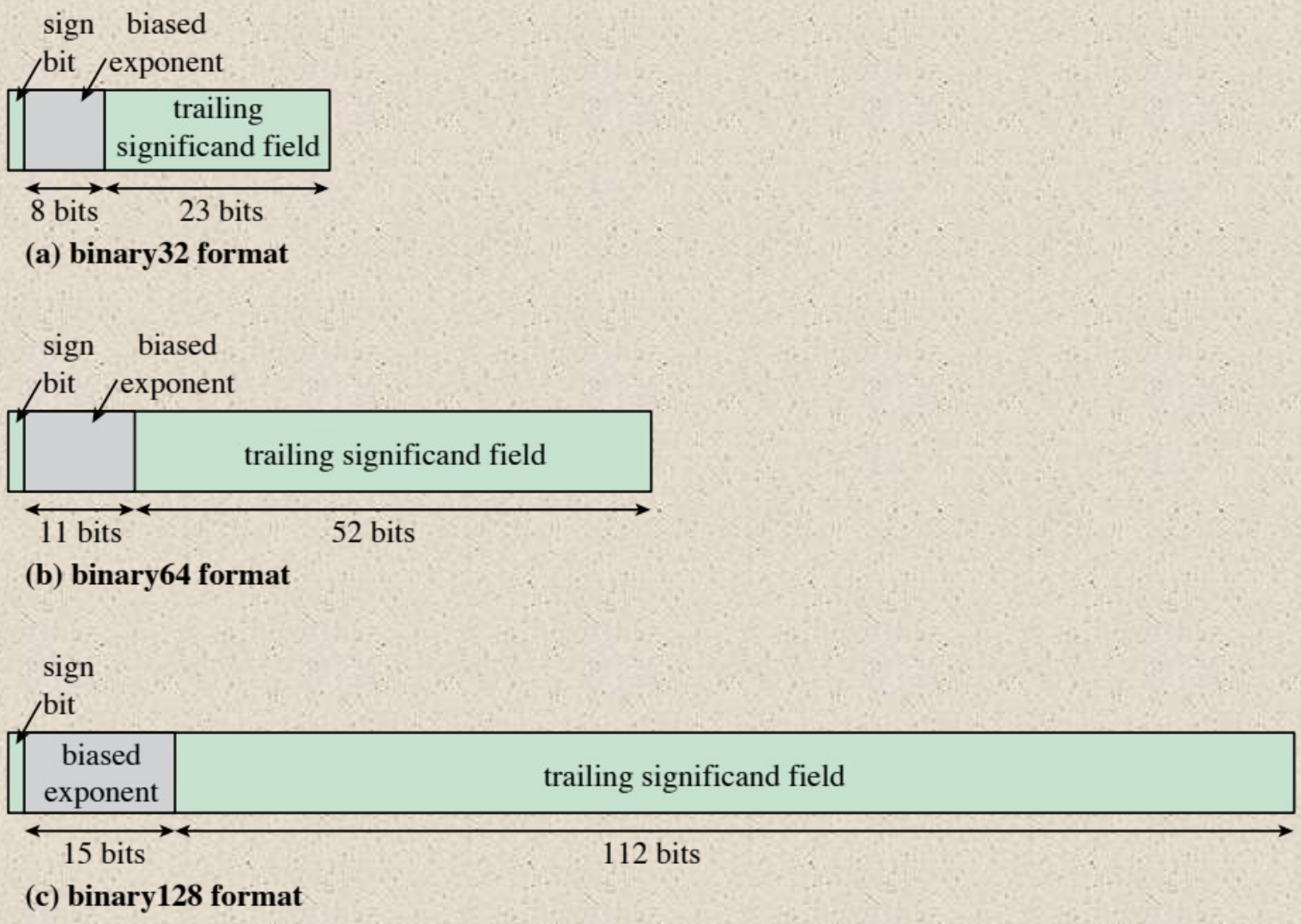

Figure 10.21 IEEE 754 Formats

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Table 10.3 IEEE 754 Format Parameters

Parameter	Format		
	binary32	**binary64**	**binary128**
Storage width (bits)	32	64	128
Exponent width (bits)	8	11	15
Exponent bias	127	1023	16383
Maximum exponent	127	1023	16383
Minimum exponent	–126	–1022	–16382
Approx normal number range (base 10)	$10_{-38}, 10_{+38}$	$10_{-308}, 10_{+308}$	$10_{-4932}, 10_{+4932}$
Trailing significand width (bits)*	23	52	112
Number of exponents	254	2046	32766
Number of fractions	2_{23}	2_{52}	2_{112}
Number of values	$1.98 \times 2_{31}$	$1.99 \times 2_{63}$	$1.99 \times 2_{128}$
Smallest positive normal number	2_{-126}	2_{-1022}	2_{-16362}
Largest positive normal number	$2_{128} - 2_{104}$	$2_{1024} - 2_{971}$	$2_{16384} - 2_{16271}$
Smallest subnormal magnitude	2_{-149}	2_{-1074}	2_{-16494}

* not including implied bit and not including sign bit

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Additional Formats

Extended Precision Formats

- Provide additional bits in the exponent (extended range) and in the significand (extended precision)
- Lessens the chance of a final result that has been contaminated by excessive roundoff error
- Lessens the chance of an intermediate overflow aborting a computation whose final result would have been representable in a basic format
- Affords some of the benefits of a larger basic format without incurring the time penalty usually associated with higher precision

Extendable Precision Format

- Precision and range are defined under user control
- May be used for intermediate calculations but the standard places no constraint or format or length

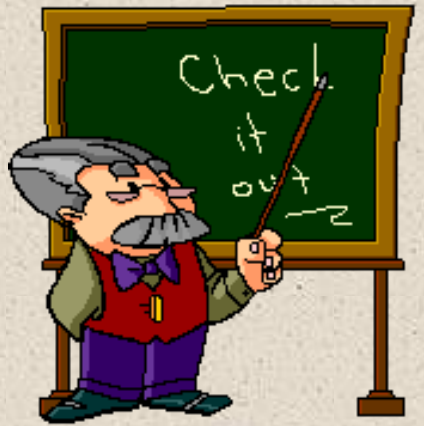

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Table 10.4
IEEE Formats

Format	Format Type		
	Arithmetic Format	Basic Format	Interchange Format
binary16			X
binary32	X	X	X
binary64	X	X	X
binary128	X	X	X
binary{k} ($k = n \times 32$ for $n > 4$)	X		X
decimal64	X	X	X
decimal128	X	X	X
decimal{k} ($k = n \times 32$ for $n > 4$)	X		X
extended precision	X		
extendable precision	X		

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Table 10.5
Interpretation of IEEE 754 Floating-Point Numbers (page 1 of 3)

	Sign	Biased exponent	Fraction	Value
positive zero	0	0	0	0
negative zero	1	0	0	–0
plus infinity	0	all 1s	0	∞
Minus infinity	1	all 1s	0	$-\infty$
quiet NaN	0 or 1	all 1s	$\neq 0$; first bit = 1	qNaN
signaling NaN	0 or 1	all 1s	$\neq 0$; first bit = 0	sNaN
positive normal nonzero	0	$0 < e < 255$	f	$2_{e-127}(1.f)$
negative normal nonzero	1	$0 < e < 255$	f	$-2_{e-127}(1.f)$
positive subnormal	0	0	$f \neq 0$	$2_{e-126}(0.f)$
negative subnormal	1	0	$f \neq 0$	$-2_{e-126}(0.f)$

(a) binary32 format

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Table 10.5
Interpretation of IEEE 754 Floating-Point Numbers (page 2 of 3)

	Sign	Biased exponent	Fraction	Value
positive zero	0	0	0	0
negative zero	1	0	0	–0
plus infinity	0	all 1s	0	∞
Minus infinity	1	all 1s	0	$-\infty$
quiet NaN	0 or 1	all 1s	$\neq 0$; first bit = 1	qNaN
signaling NaN	0 or 1	all 1s	$\neq 0$; first bit = 0	sNaN
positive normal nonzero	0	$0 < e < 2047$	f	$2_{e-1023}(1.f)$
negative normal nonzero	1	$0 < e < 2047$	f	$-2_{e-1023}(1.f)$
positive subnormal	0	0	$f \neq 0$	$2_{e-1022}(0.f)$
negative subnormal	1	0	$f \neq 0$	$-2_{e-1022}(0.f)$

(a) binary64 format

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Table 10.5
Interpretation of IEEE 754 Floating-Point Numbers (page 3 of 3)

	Sign	Biased exponent	Fraction	Value
positive zero	0	0	0	0
negative zero	1	0	0	–0
plus infinity	0	all 1s	0	∞
minus infinity	1	all 1s	0	$-\infty$
quiet NaN	0 or 1	all 1s	$\neq 0$; first bit = 1	qNaN
signaling NaN	0 or 1	all 1s	$\neq 0$; first bit = 0	sNaN
positive normal nonzero	0	all 1s	f	$2_{e-16383}(1.f)$
negative normal nonzero	1	all 1s	f	$-2_{e-16383}(1.f)$
positive subnormal	0	0	$f \neq 0$	$2_{e-16383}(0.f)$
negative subnormal	1	0	$f \neq 0$	$-2_{e-16383}(0.f)$

(a) binary128 format

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Table 10.6 Floating-Point Numbers and Arithmetic Operations

Floating Point Numbers	Arithmetic Operations
$X = X_S \times B^{X_E}$ $Y = Y_S \times B^{Y_E}$	$\left.\begin{array}{l} X + Y = \left(X_s \times B^{X_E - Y_E} + Y_s\right) \times B^{Y_E} \\ X - Y = \left(X_s \times B^{X_E - Y_E} - Y_s\right) \times B^{Y_E} \end{array}\right\} X_E \le Y_E$ $X \times Y = \left(X_s \times Y_s\right) \times B^{X_E + Y_E}$ $\frac{X}{Y} = \left(\frac{X_s}{Y_s}\right) \times B^{X_E - Y_E}$

Examples:

$X = 0.3 \times 10^2 = 30$
$Y = 0.2 \times 10^3 = 200$

$X + Y = (0.3 \times 10_{2-3} + 0.2) \times 10_3 = 0.23 \times 10_3 = 230$
$X - Y = (0.3 \times 10_{2-3} - 0.2) \times 10_3 = (-0.17) \times 10_3 = -170$
$X \times Y = (0.3 \times 0.2) \times 10_{2+3} = 0.06 \times 10_5 = 6000$
$X \div Y = (0.3 \div 0.2) \times 10_{2-3} = 1.5 \times 10_{-1} = 0.15$

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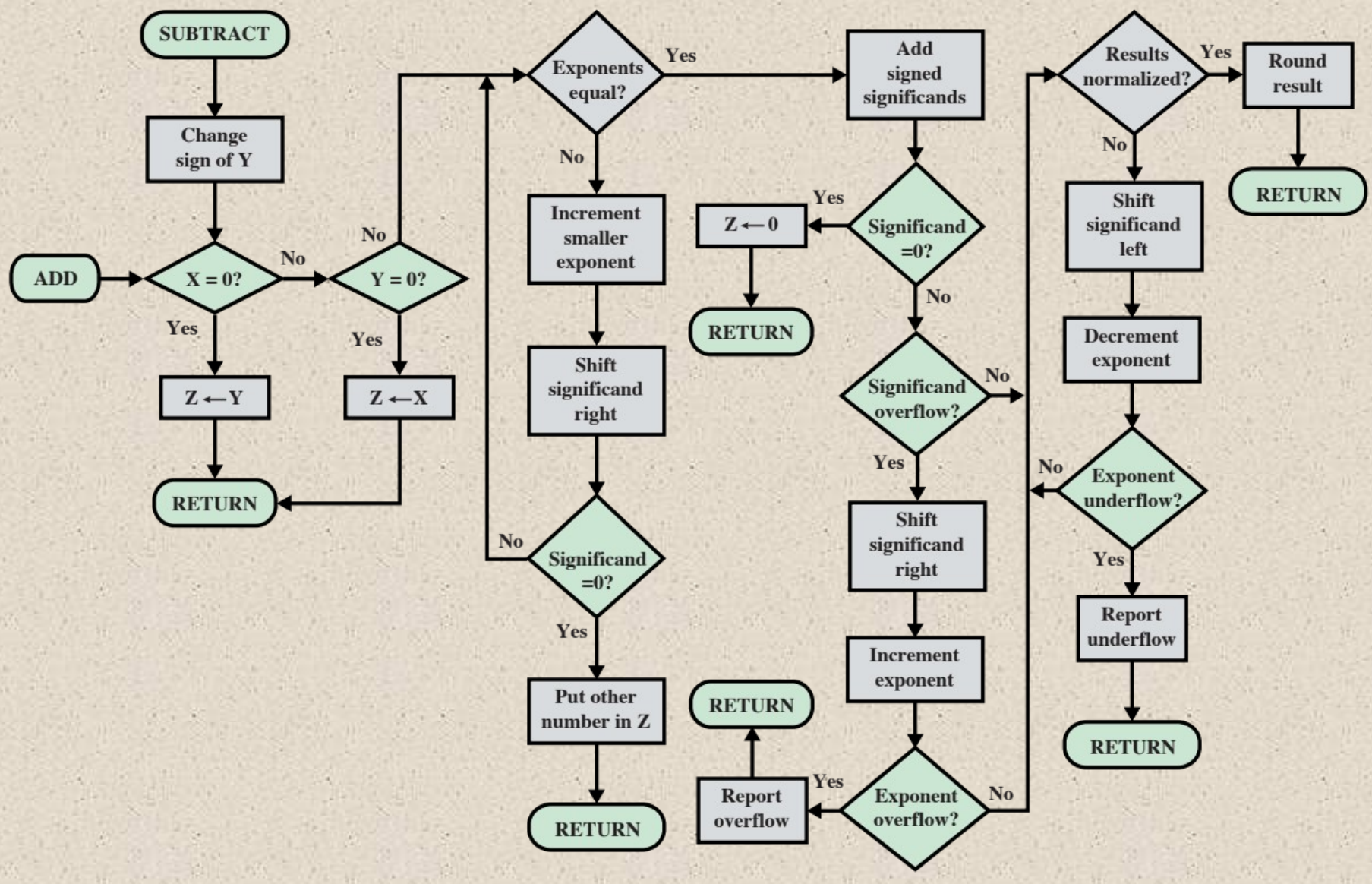

Figure 10.22 Floating-Point Addition and Subtraction (Z ← X ± Y)

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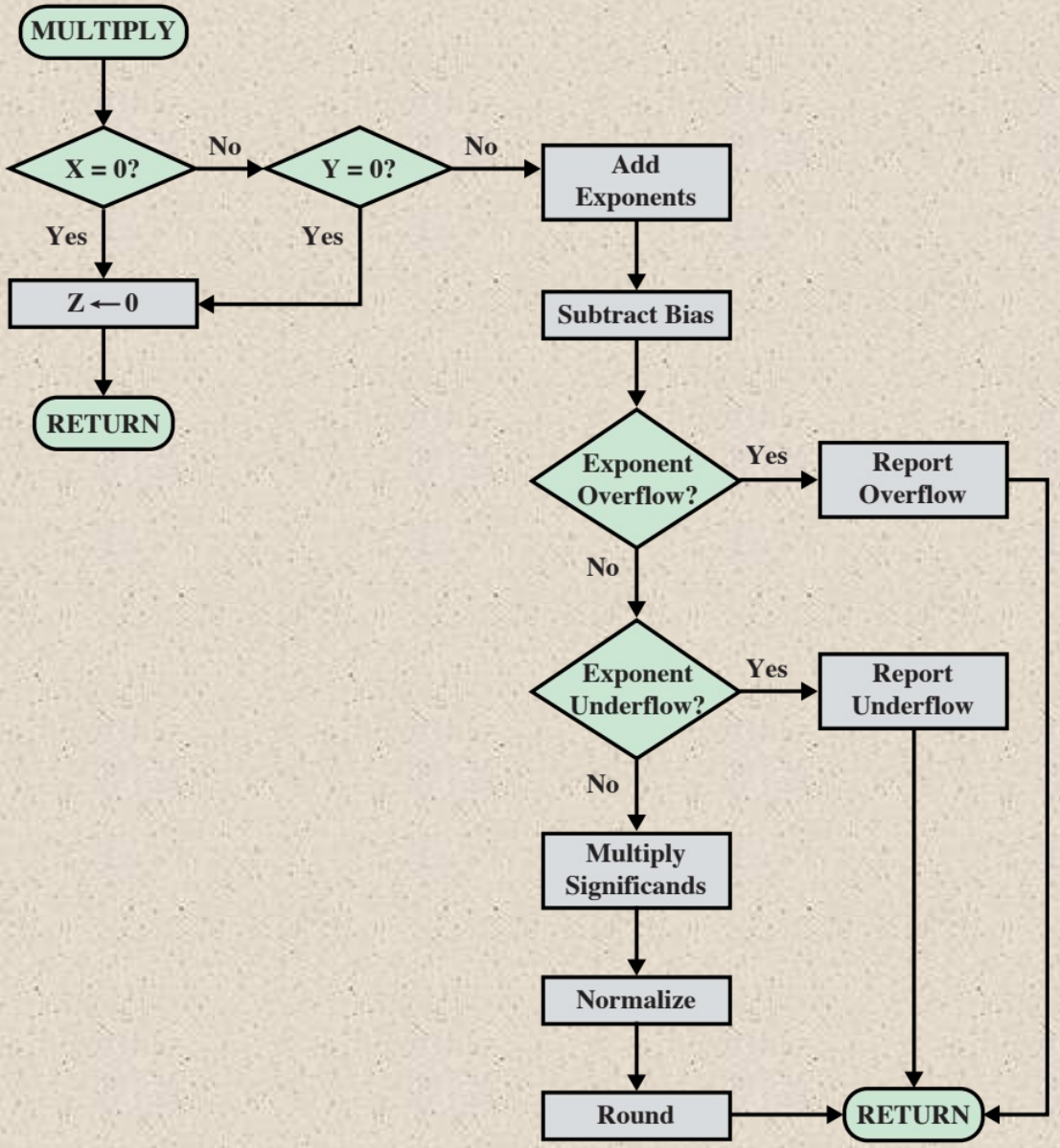

Figure 10.23 Floating-Point Multiplication (Z← X× Y)

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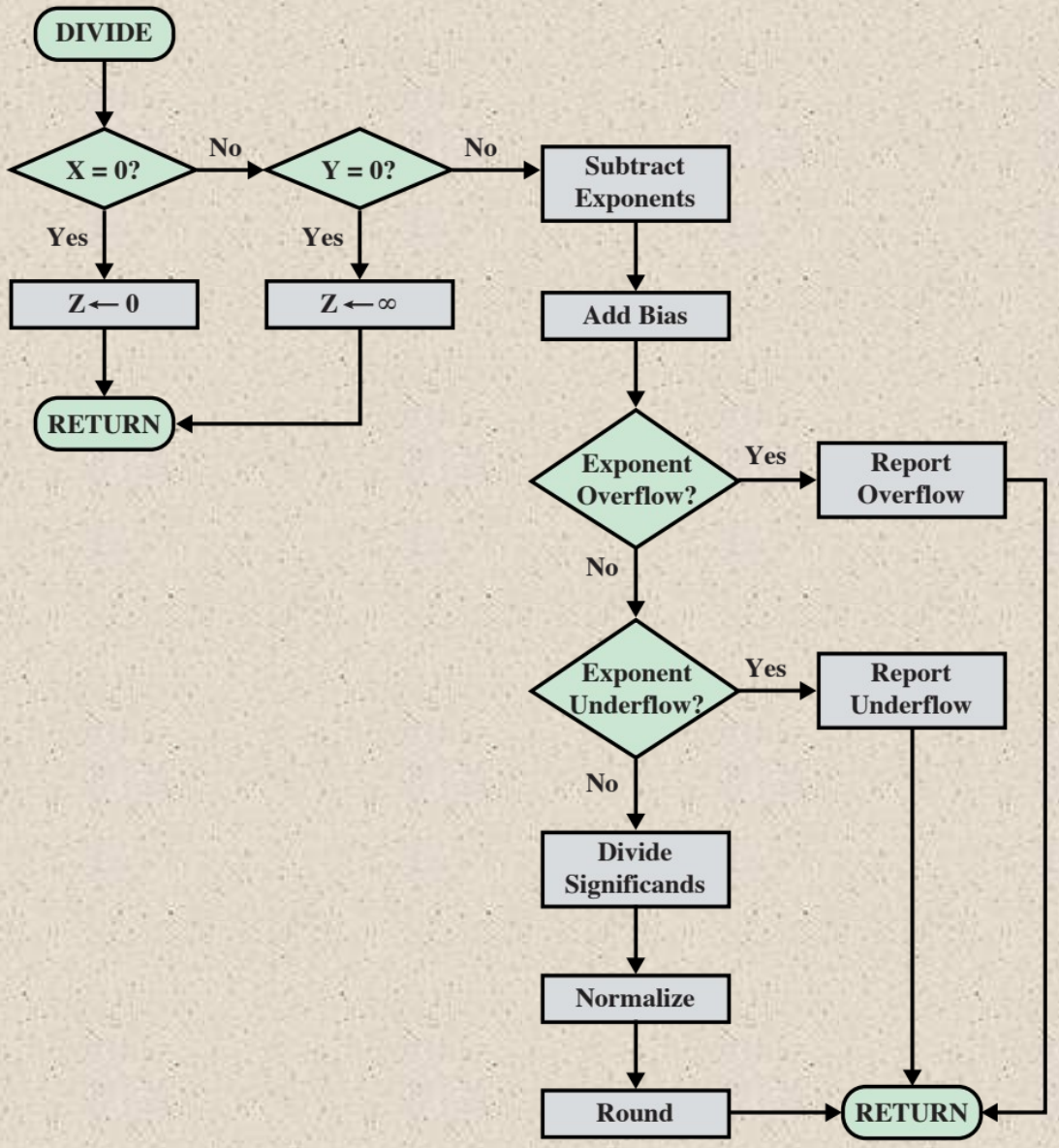

Figure 10.24 Floating-Point Division (Z← X/Y)

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$$
\begin{aligned}
x &= 1.000.....00 \times 2^{1} \\
\underline{-y} &= \underline{0.111.....11} \times 2^{1} \\
z &= 0.000.....01 \times 2^{1} \\
&= 1.000.....00 \times 2^{-22}
\end{aligned}
$$

(a) Binary example, without guard bits

$$
\begin{aligned}
x &= .100000 \times 16^{1} \\
\underline{-y} &= \underline{.0FFFFF} \times 16^{1} \\
z &= .000001 \times 16^{1} \\
&= .100000 \times 16^{-4}
\end{aligned}
$$

(c) Hexadecimal example, without guard bits

$$
\begin{aligned}
x &= 1.000.....00\ 0000 \times 2^{1} \\
\underline{-y} &= \underline{0.111.....11\ 1000} \times 2^{1} \\
z &= 0.000.....00\ 1000 \times 2^{1} \\
&= 1.000.....00\ 0000 \times 2^{-23}
\end{aligned}
$$

(b) Binary example, with guard bits

$$
\begin{aligned}
x &= .100000\ 00 \times 16^{1} \\
\underline{-y} &= \underline{.0FFFFF\ F0} \times 16^{1} \\
z &= .000000\ 10 \times 16^{1} \\
&= .100000\ 00 \times 16^{-5}
\end{aligned}
$$

(d) Hexadecimal example, with guard bits

Figure 10.25 The Use of Guard Bits

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Precision Considerations

Rounding

- IEEE standard approaches:
  - Round to nearest:
    - The result is rounded to the nearest representable number.
  - Round toward +∞ :
    - The result is rounded up toward plus infinity.
  - Round toward -∞:
    - The result is rounded down toward negative infinity.
  - Round toward 0:
    - The result is rounded toward zero.

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Interval Arithmetic

- Provides an efficient method for monitoring and controlling errors in floating-point computations by producing two values for each result
- The two values correspond to the lower and upper endpoints of an interval that contains the true result
- The width of the interval indicates the accuracy of the result
- If the endpoints are not representable then the interval endpoints are rounded down and up respectively
- If the range between the upper and lower bounds is sufficiently narrow then a sufficiently accurate result has been obtained

- *Minus infinity* and *rounding to plus* are useful in implementing interval arithmetic

Truncation

- *Round toward zero*
- Extra bits are ignored
- Simplest technique
- A consistent bias toward zero in the operation
  - Serious bias because it affects every operation for which there are nonzero extra bits

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IEEE Standard for Binary Floating-Point Arithmetic

Infinity

Is treated as the limiting case of real arithmetic, with the infinity values given the following interpretation:

$$-\infty < (\text{every finite number}) < +\infty$$

For example:

$5 + (+\infty) = +\infty$	$5 \div (+\infty) = +0$
$5 - (+\infty) = -\infty$	$(+\infty) + (+\infty) = +\infty$
$5 + (-\infty) = -\infty$	$(-\infty) + (-\infty) = -\infty$
$5 - (-\infty) = +\infty$	$(-\infty) - (+\infty) = -\infty$
$5 * (+\infty) = +\infty$	$(+\infty) - (-\infty) = +\infty$

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Table 10.7
Operations that Produce a Quiet NaN

Operation	Quiet NaN Produced by
Any	Any operation on a signaling NaN
Add or subtract	Magnitude subtraction of infinities: $(+\infty) + (-\infty)$ $(-\infty) + (+\infty)$ $(+\infty) - (+\infty)$ $(-\infty) - (-\infty)$
Multiply	$0 \times \infty$
Division	$\frac{0}{0}$ or $\frac{\infty}{\infty}$
Remainder	x REM 0 or ∞ REM y
Square root	$\sqrt{x}$ where $x < 0$

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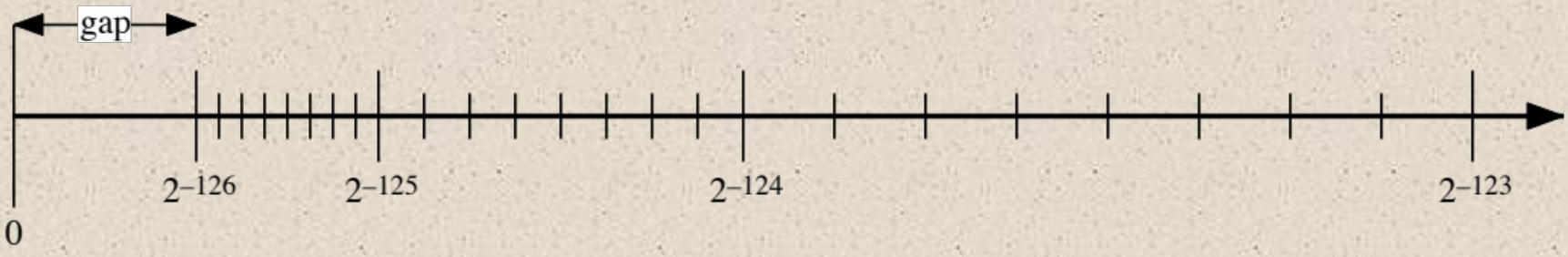
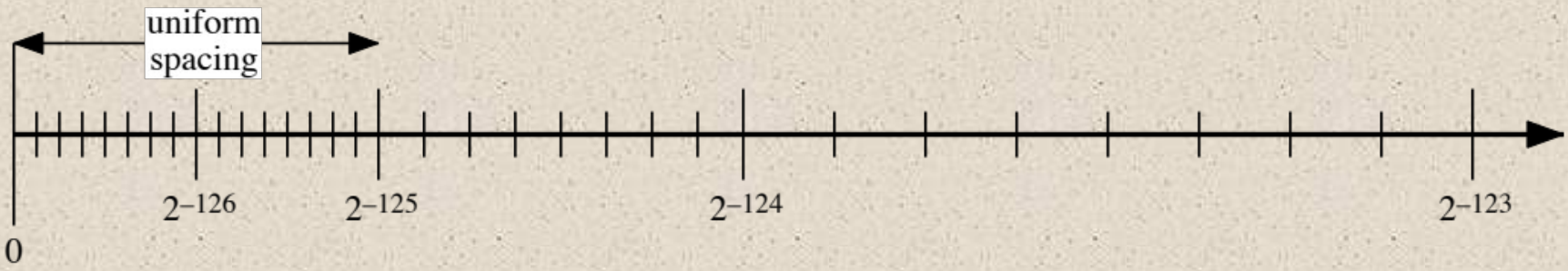


(a) 32-bit format without subnormal numbers



(b) 32-bit format with subnormal numbers

Figure 10.26 The Effect of IEEE 754 Subnormal Numbers

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Summary

Chapter 10

Computer Arithmetic

- ALU
- Integer representation
  - Sign-magnitude representation
  - Twos complement representation
  - Range extension
  - Fixed-point representation
- Floating-point representation
  - Principles
  - IEEE standard for binary floating-point representation

- Integer arithmetic
  - Negation
  - Addition and subtraction
  - Multiplication
  - Division
- Floating-point arithmetic
  - Addition and subtraction
  - Multiplication and division
  - Precision consideration
  - IEEE standard for binary floating-point arithmetic

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